

Put $e_i = 3a^2 - 2a^3$ $e_i + I^{n-1} = e_{i-1}$.

$$e_i^2 - e_i = (3a^2 - 2a^3)^2 - (3a^2 - 2a^3) = -(3-2a)(1+2a)(a^2-a)^2 = 0 \text{ in } R/I^2$$

e_i is an idem in R/I^2

if $I^n = 0$ then $e_n = f$.

Thm Nilpotent 2-side ideal of a ring R , $e \in R/I$ idem. Then there is an idem $f \in R$ lifting e .

If e is primitive, so is any lift.

Pf: Assume e primitive.

f lifts e . $f = f_1 + f_2$ idem dec.

$$e_i = f_i + I \quad e = e_1 + e_2$$

$$e_i^2 = e_i \quad e_i^2 = e_i \quad e_1 e_2 = e_2 e_1 = 0$$

So $e_1 = 0$ or $e_2 = 0 \Rightarrow f_1 \in I$ or $f_2 \in I$.

$\Rightarrow f_1 \in I$ or $f_2 \in I$ contradiction.

Since $f_2^2 = f_2$.

Corollary: Nilpotent ideal of R , Let $I = e_1 + \dots + e_n$ idem dec. in R/I . Then we have an idem dec $I = f_1 + \dots + f_n$ in R with $e_i = f_i + I$.

If e_i 's are primitive so are f_i 's.

Pf: induction on n . $n=1$ is proved.

Assume it holds for $m < n$.

Write $I = e_1 + E$ with $E = e_2 + \dots + e_n$ in R/I .

E is idem and E is orth to e_1 .

By thm e_1 is lift to f_1 , $e_1 = f_1 + I$

write $F = I - f_1 \Rightarrow F$ lifts E

F is identity element of FRF .

FIF is a nilpotent ideal FRF

$$FRF \hookrightarrow R \twoheadrightarrow R/I$$

$\hookrightarrow \varphi$ have kernel $FRF \cap I$.

Claim $FRF \cap I = FIF$

$$\supseteq x \in FIF \quad I \text{ is } \mathbb{Z}\text{-side ideal} \\ \subseteq \forall x \in FRF \cap I \quad x = Fx \in FIF \quad \text{since } F^2 x F^2 = Fx F$$

monomorphism $\frac{FRF}{FIF} \hookrightarrow R/I$

Its Image is $E(R/I)E$

In the ring $E(R/I)E$ the identity element E is a sum of m orth idem
 $E = e_1 + \dots + e_n$

$\leadsto$ the ring $\frac{FRF}{FIF}$ we get similar expansion of $F + FIF$

By induction there is an idem dec $F = f_2 + \dots + f_n$ also lifts the idem dec in E in R/I i.e. $f_i + I = e_i$

$$\forall i \geq 2 \quad F f_i = f_i \quad f_1 f_i = f_i F f_i = 0$$

$$\text{By lift } 1 = f_1 + F$$

cor I nilpotent ideal of R . $f \in R$ idem

Then f is primitive in R iff $f + I$ is primitive in R/I

Recall $f \in R$ idem. f is the identity elt. of fRf .
 If $f = f_1 + f_2$ is an idem dec in R then $f_1, f_2 \in fRf$.

$$\text{Pf: } f f_1 = f_1 \quad f_1 \in fRf$$

Lemma Let A be semisimple Artinian ring.

(1) There is a primitive idem dec $1 = e_1 + \dots + e_n$.

This corresponds to $A = Ae_1 \oplus \dots \oplus Ae_n$ where Ae_i is simple.

(2) For any simple A -mod S there is a primitive idem e such that $S \cong Ae$

(3) For any simple A -mod S there is $1 \leq i \leq n$ st $e_i S \neq 0$ and if T is a simple A -mod with $T \not\cong S$ then $e_i T = 0$

Pf (1) By decomposition of module dec.

(2) S is a direct summand of A

(3) $S = 1 \cdot S = (e_1 + \dots + e_n)S$

$\Rightarrow \exists i$ s.t. $e_i S \neq 0$.

$e_i A \cong \text{Hom}_R(Ae, Ae_i)$

follows from Schur's lemma.

Thm. A Artinian ring S simple A -mod.

(1) There is an indecomposable projective A -module P_S with $P_S/\text{Rad}(P_S) \cong S$ of the form $P_S = Af$ where f is a primitive idem in A .

(2) The idem f has the property that $fS \neq 0$ and if T is any simple mod. with $T \not\cong S$ then $fT = 0$.

(3) P_S is the proj. cover of S .

Pf. Let $e \in J(A) \Rightarrow J(A)$ nilpotent. be any primitive idem with $eS \neq 0$

Let f be any lift of e to A . $e = f + J(A)$.

f primitive. $fS = eS \neq 0$.

$fT = 0$ for any simple A -mod. $T \not\cong S$.

Set $P_S = Af$ indecomposable projective A -mod.

$Af \rightarrow (A/J(A)) \cdot (f + J(A)) = S$.

$a \mapsto (a + J(A))(f + J(A))$

with kernel $J(A) \cdot Af = \text{Rad}(Af)$

So $P_S/\text{Rad}(P_S) \cong S$.

P_S is the project cover of S .

Thm. A : Artinian ring, Up to isomorphism the indec proj A -mods are exactly the modules P_S that are the projective cover of the simple modules and $P_S \cong P_T$ iff $S \cong T$. Each P_S appears as a direct summand of A with multiplicity equal to the multiplicity of S as a summand of $A/J(A)$.

The left regular A -module A decomposes as $A \cong \bigoplus_S (P_S)^{n_S}$ where S run through the simple A -mods up to iso. $n_S = \dim_D S$ with $D = \text{End}_A(S)$

If A is a f.d. k -algebra over an algebraically closed field k then $n_S = \dim_k S$.

Pf: f.g. indec proj. mod.

$$\text{Rad}(P) \cong S_1 \oplus \dots \oplus S_t$$

P is the proj. cover. of $S_1 \oplus \dots \oplus S_t$

but $P_{S_1} \oplus \dots \oplus P_{S_t}$ is a project cover. of $S_1 \oplus \dots \oplus S_t$

$$P = P_{S_1} \oplus \dots \oplus P_{S_t} \Rightarrow t=1$$

Cor. Let A be an Artinian Ring. U.f.g. A -mod.

Then U has a proj. cover.

Pf: $U/\text{Rad}(U) \cong S_1 \oplus \dots \oplus S_n$

$P_{S_1} \oplus \dots \oplus P_{S_n} \rightarrow S_1 \oplus \dots \oplus S_n$ proj. cover.

$$\begin{array}{ccc} & \delta & \\ & \swarrow & \\ U & \xrightarrow{g} & S_1 \oplus \dots \oplus S_n \rightarrow 0 \\ & \searrow & \downarrow R \\ & & P_{S_1} \oplus \dots \oplus P_{S_n} \end{array}$$

essential.

$P_{S_1} \oplus \dots \oplus P_{S_n}$ is a proj. cover of U

Lem 1) A semisimple. Then A is Artinian. iff A is Noetherian.

(2) A is Artinian, then A is Noetherian.

(direct) left Artinian $\Rightarrow$ left

A : Artinian ring V : f.g A -mod.

Then V has a composition series.

For every simple A -module S , the number of the composition factors isomorphic to S is called the multiplicity of S in V .

Thm. A : Artinian, $f \in A$ primitive idem. $e = f + J(A)$ primitive idem. in $A/J(A)$.

Then the multiplicity of $(A/J(A))e$ in any f.g. A -module V is equal to the composition length of fV as an fA -module.

Pf: Let $V = V_0 \supseteq V_1 \supseteq \dots \supseteq V_n = 0$ be a composition series of V , consider the series of fA -modules

$$fV = fV_0 \supseteq fV_1 \supseteq \dots \supseteq fV_n = 0$$

$$V_i/V_{i+1} \cong (A/J(A))f \Leftrightarrow f(V_i/V_{i+1}) \neq 0$$

in this case.

$$e(A/J(A))e \simeq f(V_i/V_{i+1})f(V_i/V_{i+1})/V_{i+1} \simeq J^i V_i / J^i V_{i+1} = J^i V_i / J^i V_{i+1} \text{ which is simple } fA\text{-module.}$$

Let $I = e_1 + \dots + e_n$ primitive idem dec in $A/J(A)$.

$$I = f_1 + \dots + f_n \text{ in } A \text{ with } e_i = f_i + J(A)$$

We denote by c_{ij} the multiplicity of $(A/J(A))e_i$ in $f_j A$

which is called the Cartan invariant of A

The $n \times m$ matrix (c_{ij}) is called the Cartan matrix of A .

$$\text{Thm: } c_{ij} \neq 0 \Leftrightarrow f_i A f_j \neq 0.$$

Prop: A f.d. k -alg. over a field k . S simple A -mod. with proj. cover P_S . U f.d. A -mod.

$$(1) \text{ If } T \text{ is a simple } A\text{-mod. then } \dim_k \text{Hom}_A(P_S, T) = \begin{cases} \dim_k \text{End}_A(S) & \text{if } S \simeq T \\ 0 & \text{otherwise.} \end{cases}$$

$$(2) \text{ The multiplicity of } S \text{ in } U \text{ is } \dim_k \text{Hom}(P_S, U) / \dim_k \text{End}_A(S)$$

$$(3) e \in A \text{ is an idem. Then } \dim_k \text{Hom}(Ae, U) = \dim_k eU$$

$$(4) \text{ Let } e_s, e_t \text{ be idempotents so that } P_S = Ae_s, P_T = Ae_t \text{ projective covers of } S \text{ and } T \\ \text{Then } c_{ST} = \dim_k \text{Hom}_A(P_S, P_T) / \dim_k \text{End}_A(S) = \dim_k e_s A e_t$$

$$\text{Pf (1)} \text{ Hom}_A(P_S, T) : P_S \rightarrow T. \text{ nonzero. im}(f) = T$$

$$f : P_S \rightarrow T \text{ ker } f \text{ is a maximal submodule. } \text{Rad}(P_S) \subseteq \text{ker } f.$$

f factor through $P_S / \text{Rad}(P_S)$.

$$P_S \rightarrow P_S / \text{Rad } P_S \rightarrow T \simeq \text{Hom}_A(P_S, T) \simeq \text{Hom}(S, T)$$

Integrality

Def. S = integral domain. R subring of S containing 1. Let $\alpha \in S$. We say α is an integral element over R if there exists a monic polynomial $f \in R[X]$ s.t. $f(\alpha) = 0$.

Prop: S, R as above. $\alpha \in S$. $T = R[\alpha]$.

(1) α is integral over R .

(2) $R[\alpha]$ is a f.g. R -module.

(3) There is a subring T of S with $R \subseteq T$ s.t. T is f.g. as an R -module and $\alpha \in T$.

Pf: (2) $\Rightarrow$ (3) Let $T = R[\alpha]$

(1) $\Rightarrow$ (2). α integral. $\exists$ monic $f \in R[X]$. $f(X) = X^n + a_{n-1}X^{n-1} + \dots + a_1X + a_0$ s.t. $f(\alpha) = \alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_0 = 0$
 $\alpha^n = -a_{n-1}\alpha^{n-1} - \dots - a_1\alpha - a_0$

$\Rightarrow$ every power of α is an R -linear combination of $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$.

$\Rightarrow$ as an R -mod, $R[\alpha]$ is generated by $1, \alpha, \dots, \alpha^{n-1}$

(3) $\Rightarrow$ (1) Let $\varepsilon_1, \dots, \varepsilon_n$ be the set of generators of T over R .

Write $\alpha \varepsilon_i = a_{i1}\varepsilon_1 + \dots + a_{in}\varepsilon_n$. $a_{ij} \in R$.

$$\Rightarrow (\varepsilon_1, \dots, \varepsilon_n) \begin{pmatrix} \alpha a_{11} & \alpha a_{12} & \dots & \alpha a_{1n} \\ -a_{12} & \alpha a_{22} & \dots & -a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n2} & \dots & \dots & \alpha a_{nn} \end{pmatrix} = 0$$

$$= A$$

$$\text{multiplying } A^* \Rightarrow (\varepsilon_1, \dots, \varepsilon_n) \begin{pmatrix} \det A & & \\ & \ddots & \\ & & \det A \end{pmatrix} = 0$$

$$\varepsilon_i (\det A) = 0.$$

$$\Rightarrow (\det A) \cdot T = 0$$

$$\det A = 0.$$

$\det A$ is a monic poly over R view α as X .

Lem S integral domain. R subring of S with $1_S \in R$

$R \subseteq M_1 \subseteq S$ where M_i is a subring of S . ($i=1,2$) If M_1 and M_2 are f.g. as R -mods, then the ring $R[M_1, M_2]$ generated by R, M_1, M_2 is f.g. as an R -module.

Pf: M_1 generators $\alpha_1, \dots, \alpha_m$ as R -mod.

M_2 generators $\beta_1, \dots, \beta_n$ as R -mod.

Then $\{\alpha_i \beta_j \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ generates $R[M_1, M_2]$

Thm: S, R as above. Then the integral elts of S over R form a subring of S containing R .

Pf. α, β integral over $R \Rightarrow \alpha + \beta, \alpha\beta$ also integral over R .

$R[\alpha, \beta]$ is f.g. over R .

$\alpha + \beta, \alpha\beta \in R[\alpha, \beta]$.

So $\alpha + \beta, \alpha\beta$ are integral.

Def. $\alpha \in \mathbb{C}$, $\mathbb{C}$ is integral over $\mathbb{Z}$, we call α an algebraic integer.

Def. S, R as above. R is called integrally closed in S if all integral elts of S over R lie in R .

Thm. A UFD is integrally closed.

Pf. $R = \text{UFD}$. K : quotient field. Let $\alpha \in K$ be integral over R .

write $\alpha = \frac{b}{a}$ with $\gcd(a, b) = 1$.

There exists $n \geq 1, c_1, \dots, c_n \in R$ s.t. $\left(\frac{b}{a}\right)^n + c_{n-1}\left(\frac{b}{a}\right)^{n-1} + \dots + c_0 = 0$

$$b^n + (c_{n-1}b^{n-1} + \dots + c_0a^{n-1})a = 0.$$

If a is not a unit, any prime divisor of a divides b contra $\Rightarrow a$ is a unit $\Rightarrow \alpha \in R$.

Valuation.

K : field. An valuation of K is a real-valued function v defined on $K^\times = K \setminus \{0\}$ satisfying $v(ab) = v(a) + v(b)$ and $v(ab) \geq \min\{v(a), v(b)\}$.

Prop. $v: (K^\times, \cdot) \rightarrow (\mathbb{R}, +)$ group homo.

Set $v(0) = \infty$

$v(K^\times)$ is subgroup of $(\mathbb{R}, +)$ called value group of v .

A field K with valuation v , denote (K, v) , called valuation field.

Prop. (1) $v(\pm 1) = 0$. (2) $v(a) = v(-a)$. (3) $v(a^{-1}) = -v(a)$. (4) $v(b) < v(a) \Rightarrow v(ab) = v(b)$.